# Citric Acid Production

# Process Modeling and Evaluation with SuperPro Designer®

by

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This is the ReadMe file of a SuperPro Designer example that deals with process modeling, cost analysis and optimization of Citric Acid production via fermentation. The flowsheet of the process is appended to the bottom of this document. You may test-drive the model by downloading the functional evaluation edition of SuperPro Designer from the downloads page of our website (**www.intelligen.com**). The files of this example can be found in the **Examples \ Bio-Materials \ CitricAcid** folder. The default installation path of the SuperPro Designer Examples folder follows below.

C:\ Users \ Public \ Public Documents \ Intelligen \ SuperPro Designer \ v11 \ Process Library \ Examples

If you have any questions about this example and SuperPro Designer in general, please send an email message to dpetrides@intelligen.com

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## Introduction

This example analyzes the production of citric acid, which is a commodity organic acid used in the food and beverage industries to preserve and enhance flavor [1]. In this example citric acid is produced via fermentation using *Aspergillus niger.* The plant considered here produces around 18,000 metric tons of crystal citric acid per year, which represents approximately 1% of the current world demand.

For reporting purposes, the process has been divided into two sections:

- The Fermentation Section
- The Isolation Section

Flowsheet sections in SuperPro are simply sets of related unit procedures (processing steps). The unit procedure icons of the Fermentation section are displayed in green, while the icons of the Isolation section are displayed in blue. For information on how to specify flowsheet sections and edit their properties, please consult Chapter 8.1 of the SuperPro manual.

## Process Description

**Fermentation Section -** The flowsheet of the entire process is appended to the bottom of this document. Molasses, the carbon source for this fermentation, is diluted with water from about 50% fermentable sugar content to 20% fermentable sugar content in a blending tank (V-101). Suspended particulate material is then removed by filtration (PFF-101). Metal ions, particularly iron, are subsequently removed by an ion exchange chromatography column (C-101) and the purified raw material solution is then heat-sterilized (ST-101). Nutrients (i.e., sources of ammonium, potassium, phosphorus, magnesium, copper, and zinc) are dissolved in water (V-104) and heat-sterilized (ST-101).

After sterilization, both the carbon source and the nutrients are transferred into the fermentation vessel (FR-101). A three-step seed fermentor train, not shown in the flowsheet, supplies inoculum to the fermentor. Fermentation is then carried out according to the following mass stoichiometry:

**2 (Ammonium Sulfate) + 180 (Glucose) + 12 (Nutrients) + 56 (Oxygen) ➔**

**16 (Biomass) + 30 (Carbon Dioxide) + 154 (Citric Acid) + 50 (Water)**

The extent of the fermentation reaction is set to 99% and the fermentation is carried out for a period of 5 days at a temperature of 35℃. Air is supplied by a compressor (G-101) at a rate that gradually increases from 0.15 VVM (volume of air per volume of liquid per minute) to 1.0 VVM. Cooling water removes the heat produced by the exothermic process and maintains a constant temperature.

To meet the production demand, fermentation is carried out by seven fermentors that operate in staggered mode (out of phase); staggered mode of operation will be explained later in this document. Moreover, the plant operates around the clock and one fermentation batch is initiated daily and another one is completed daily. Each fermentor has a vessel volume of 350 $m^3$ and generates broth of around 315

$m^3$. Once fermentation is completed, the broth is discharged into the holding tank (V-105), which acts as a buffer tank between the batch upstream section and the continuous downstream section.

**Isolation Section:** Purification starts with the removal of biomass by a rotary vacuum filter (RVF-101). The clarified fermentation liquor subsequently flows into an agitated reactor vessel (V-106) where approximately 1 part of hydrated lime, $Ca(OH)_2$, for every 3 parts of liquor is slowly added in order to precipitate calcium citrate. The lime solution must be very low in magnesium content if losses due to the relatively soluble magnesium citrate are to be avoided. Calcium citrate is then separated by a second rotary vacuum filter (RVF-102) and the citrate-free filtrate (Aq-Waste-1) is sent to a wastewater collection tank. The calcium citrate cake is sent to another agitated reactor vessel (V-107), where it is acidified with dilute sulfuric acid to form a precipitate of calcium sulfate (gypsum). A third filter (RVF-103) removes the precipitated gypsum and yields an impure citric acid solution. Careful control of pH and temperature of the precipitation steps is important for maximizing the yield of citric acid. The resulting solution is concentrated and crystallized using a continuous evaporator/crystallizer (CR-101). The crystals are then separated by filtration (RVF-104) and dried in a rotary dryer (RDR-101). If the final product is required in high purity (e.g., for pharmaceutical applications), treatment with activated carbon may precede crystallization in order to remove colorants. In addition, ion exchange is sometimes used to remove metal ions and other ionic species.

## Process Scheduling

The Equipment Occupancy Chart (EOC) for 14 consecutive batches is displayed in Figure 1. The process batch time is approximately 200 h (or 8.3 days). This is the time elapsed from the preparation of raw materials to the generation of the final product for a single batch (excluding the time required for inoculum preparation). The duration of each fermentation batch is 160 h (6.7 days). The equipment occupancy chart is a valuable tool for identifying scheduling bottlenecks (the equipment with the longest occupancy time) and performing appropriate adjustments in order to reduce the cycle time of a batch process and increase its throughput.

The availability of 7 production fermentors, operating in staggered mode (out of phase), enables the plant to initiate a new batch every 24 h. The upstream portion of the process (i.e., raw material preparation and fermentation) operates in batch mode. The downstream section (product recovery and purification) operates continuously. The continuous units of the purification train are displayed in Figure 1 with aligned blocks that operate back-to-back (lower portion of the chart). Specification of units that operate continuously in a batch flowsheet will be discussed later in this document.

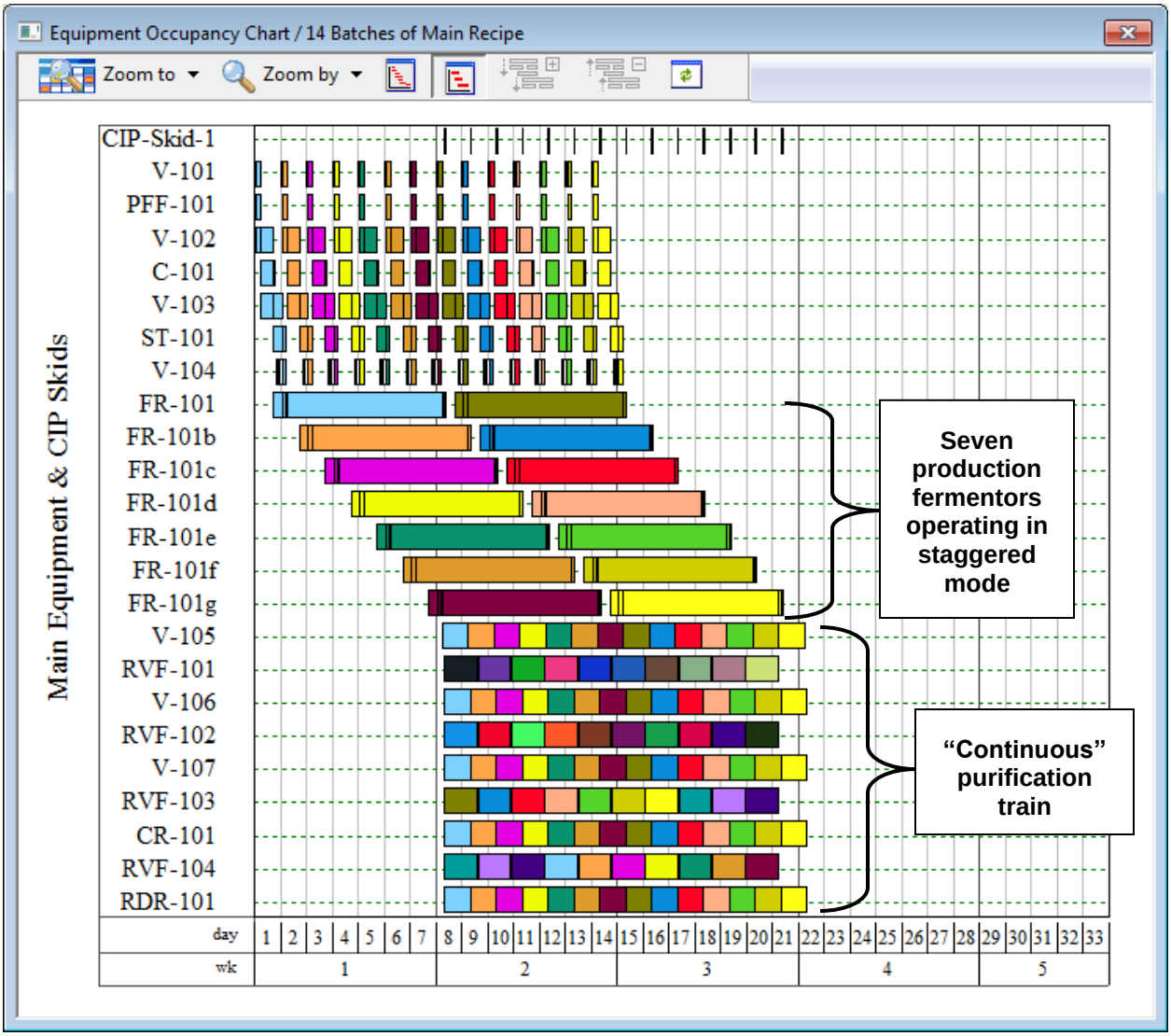



**Figure 1** Equipment occupancy chart for 14 consecutive batches of the citric acid process.

Figure 2 is the Operations Gantt chart (in the MS Project style) that displays detailed scheduling information for one or multiple batches. This chart is generated by selecting **Charts➲Gantt Charts➲Operations GC**. Figure 2 displays an extract of the full Gantt chart, illustrating the scheduling of operations in procedures P-1 to P-7. The brown bar at the top represents the duration of the entire recipe while the dark blue and cyan bars represent the durations of procedures and operations, respectively.

The Gantt chart enables users to visualize the execution of a batch process in detail. It also facilitates editing of batch recipes. Double clicking on any of its bars brings up the dialog of the corresponding entity (e.g., operation, procedure, recipe, etc.). The simulation calculations can be redone and the chart can be updated by clicking on the refresh button of the chart ( ).

SuperPro can export its scheduling data to **MS Project** by selecting **File➲Export to MS Project XML File.** Please **c**onsult the SuperPro manual or its Help facility for information on the two exporting options. Likewise, SuperPro can export its recipe data to SchedulePro by selecting **File➲Export to SchedulePro's Recipe DB.** SchedulePro is a resource management, production planning and scheduling tool marketed by Intelligen.

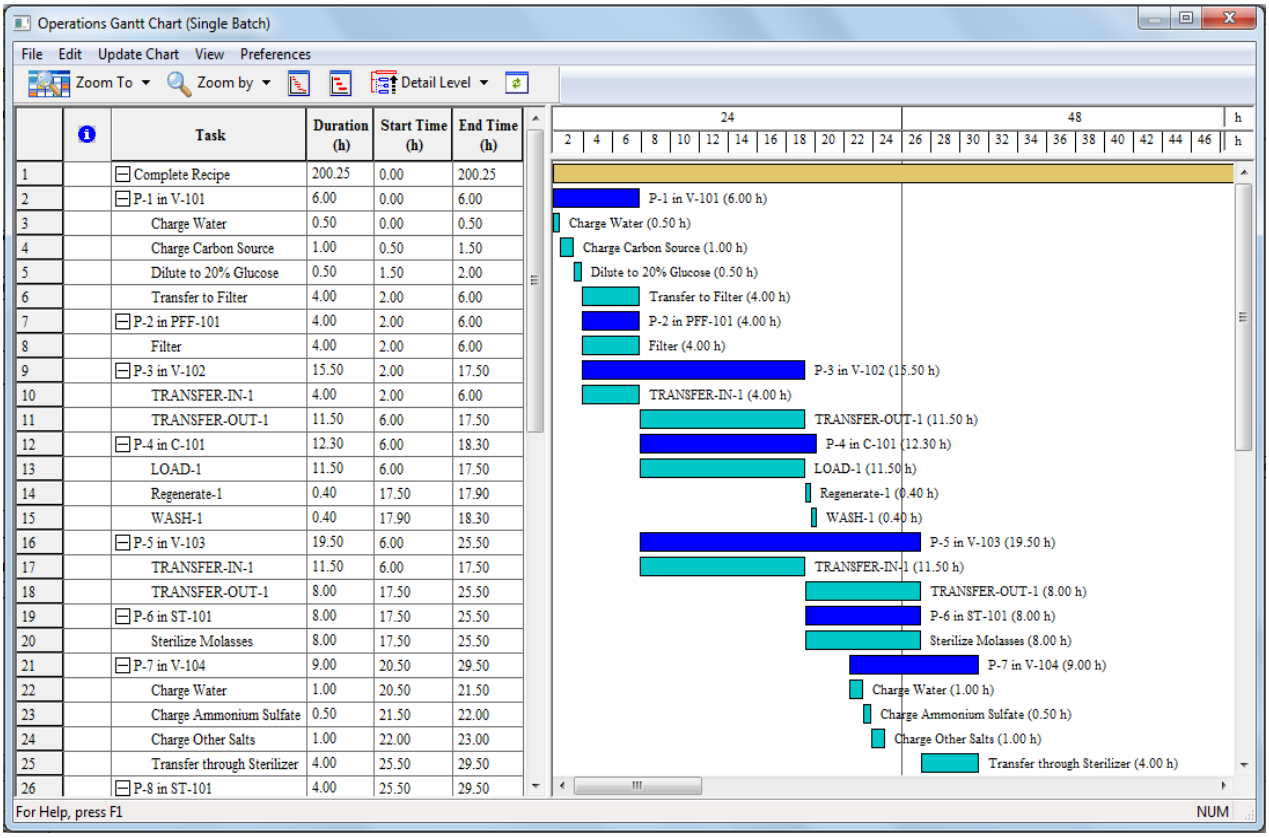



**Figure 2** Operations Gantt chart.

## Specifying Equipment in Staggered Mode

As mentioned in the process description section, fermentation is carried out by 7 fermentors that operate in staggered mode (out of phase) and feed a single purification train that operates continuously. The duration of the fermentation procedure is 6.7 days. The availability of 7 fermentors operating in staggered mode enables the plant to start one fermentor daily (and harvest one fermentor daily as well). The equipment of the purification train, which operates continuously, is sized to process the contents of one fermentor per day.

The addition of extra equipment items that operate in staggered mode is done through the equipment data dialog of a procedure, which is accessed by right-clicking on the procedure icon and selecting

**Equipment Data**. Figure 3 displays the equipment data dialog of the fermentor (FR-101). The **Stagger Mode** frame is located on the lower left corner of the dialog. Check the **On** box of that frame and specify the extra pieces of equipment that operate in staggered mode (six in this case). The names of the extra equipment items can be edited by clicking on the **Names** button in the same frame. Notice that the total number of equipment units available to the procedure is equal to the number of units in stagger mode, plus one.

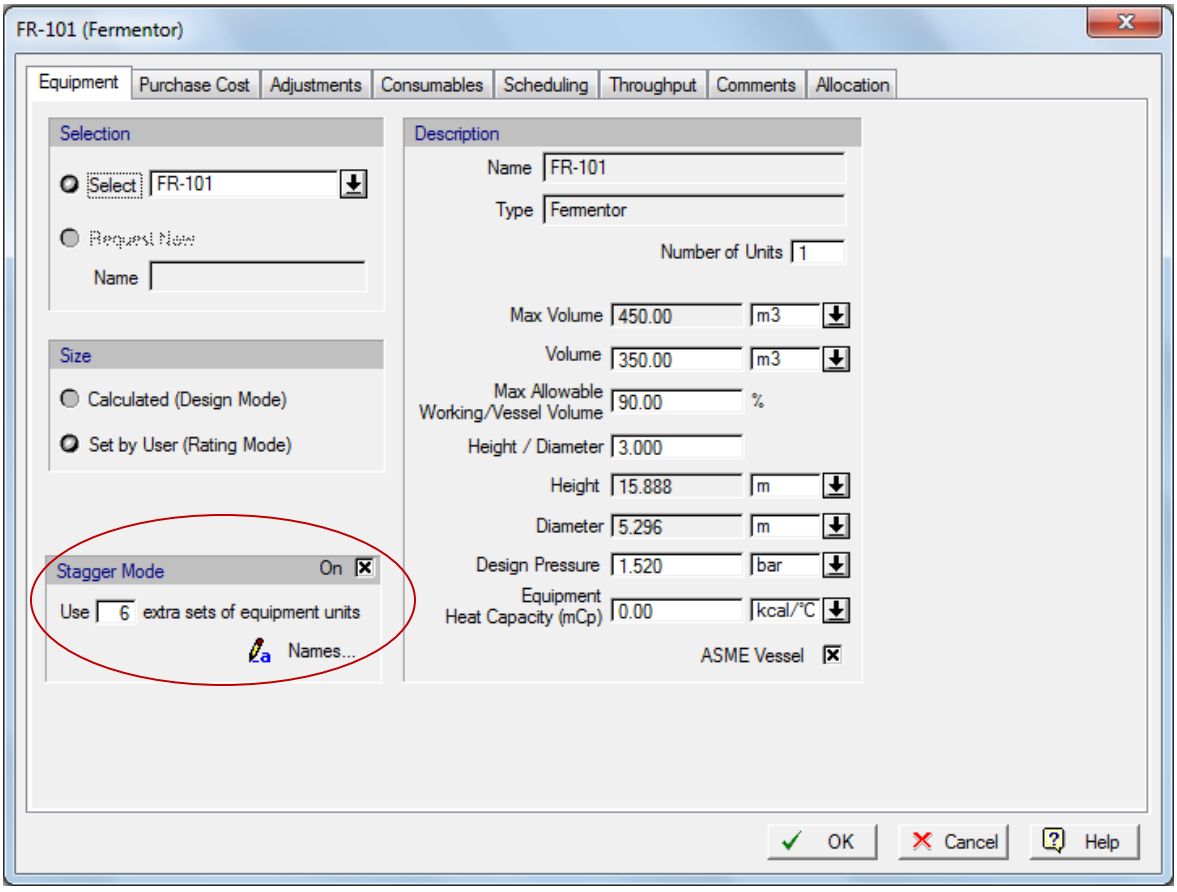



**Figure 3** Equipment Data Dialog for Fermentor FR-101.

## Material Requirements

Table 1 displays the raw material requirements in kg/yr and kg/batch, and kg/kg MP ("MP" = main product; citric acid in this case). The output of a single batch is approximately 55 metric tons of Citric Acid crystals and the total annual production volume is approximately 18,000 metric tons, as mentioned earlier. The table below was extracted from the RTF version of the **Materials & Streams** report, which can be generated by selecting **Reports \ Materials & Streams** from the main menu bar of SuperPro. The

format of the report can be specified through the dialog that is displayed when you select **Reports \ Options** from the main menu bar of SuperPro.

**Table 1** Material Requirements for the Entire Process

| BULK MATERIALS (Entire Process) | | | |
|---|---|---|---|
| **Material** | **MT/yr** | **MT/batch** | **MT/MT MP** |
| Water | 132,735 | 403.449 | 7.363 |
| Molasses | 46,008 | 139.841 | 2.552 |
| Amm. Sulfate | 278 | 0.846 | 0.015 |
| Nutrients | 1,900 | 5.775 | 0.105 |
| Nitrogen | 49 | 0.150 | 0.003 |
| NaOH (1 M) | 4,844 | 14.723 | 0.269 |
| Air | 279,813 | 850.496 | 15.521 |
| Lime (33%) | 36,304 | 110.347 | 2.014 |
| H2SO4 (10% w/w) | 153,163 | 465.542 | 8.496 |
| TOTAL | 655,094 | 1,991.168 | 36.338 |

## Cost Analysis

SuperPro Designer performs thorough cost analysis, estimating capital (CAPEX) as well as operating (OPEX) costs and generates the following three pertinent reports (through the **Reports** menu): *Economic Evaluation Report (EER)*, *Cash Flow Analysis Report (CFR),* and *Itemized Cost Report (ICR)*. Table 1 displays the **Executive Summary** of the Economic Evaluation Report. Reports can be generated in various formats (e.g., PDF, RTF, XLS, etc.) specified through **Reports➲Options.** The total capital investment for a plant of this capacity (18,000 metric tons of citric acid crystals per year) is around $50.1 million while the estimated annual operating cost is $25.9 million. The numbers for the calculation of the Return-On-Investment, Payback Time, etc. are based on a selling price of $2/kg of purified citric acid crystals.

**Table 2** Executive Summary

| EXECUTIVE SUMMARY (2013 prices) | |
|---|---|
| Total Capital Investment | 50,096,000 $ |
| Capital Investment Charged to This Project | 50,096,000 $ |
| Operating Cost | 25,909,000 $/yr |
| Revenues | 36,056,000 $/yr |
| Cost Basis Annual Rate | 18,028 MT MP/yr |
| Unit Production Cost | 1,437.14 $/MT MP |
| Net Unit Production Cost | 1,437.14 $/MT MP |
| Unit Production Revenue | 2,000.00 $/MT MP |
| Gross Margin | 28.14 % |
| Return On Investment | 17.99 % |
| Payback Time | 5.56 years |
| IRR (After Taxes) | 14.92 % |
| NPV (at 7.0% Interest) | 32,742,000 $ |
| MP = Total Flow of Stream 'Product' | |

Figure **4** displays the annual operating cost breakdown, which is part of the EER. This type of chart can be included in the report by selecting **Reports➲Options** and activating the **Include Charts** option on the lower right corner of the dialog. As can be seen in the chart, the raw materials cost is the most important, accounting for 40% of the overall operating cost. This is quite common for commodity bio-chemicals. Molasses is the most expensive raw material, accounting for 66% of the raw materials cost. The purification chemicals, sulfuric acid and calcium hydroxide, account for 19% and 9% of the overall raw materials cost, respectively. The following material prices were assumed for this analysis: $0.15/kg of molasses, $0.013/kg of 10% w/w H2SO4 solution, $0.08/kg of Ca(OH)2, and $0.001/kg of process water.

The facility-dependent expense is the second most important operating cost component, accounting for 30% of the overall cost. Depreciation of the fixed capital investment and maintenance of the facility are the main contributors to this cost. Utilities are the third largest expense, accounting for 14% of the overall cost. Electricity and cooling water utilized by the fermentors are the main contributors to this cost. Labor lies in the fourth position, and the environmental cost (waste treatment/disposal) is fifth. Disposal unit costs of $1/m$^3$ and $50/MT (metric ton) were assumed for liquid and solid (gypsum) waste streams, respectively. The disposal of gypsum accounts for 85% of the overall environmental cost. The overall unit production cost is approximately $1.4/kg, which is roughly equal to the current (early 2012) selling price of citric acid [2]. This can be explained by noting the excess capacity around the world and the fact that most operating citric acid plants are rather old and partially depreciated. If depreciation is ignored, the facility-dependent cost is reduced by more than 80% and the overall unit cost drops to around $1/kg.

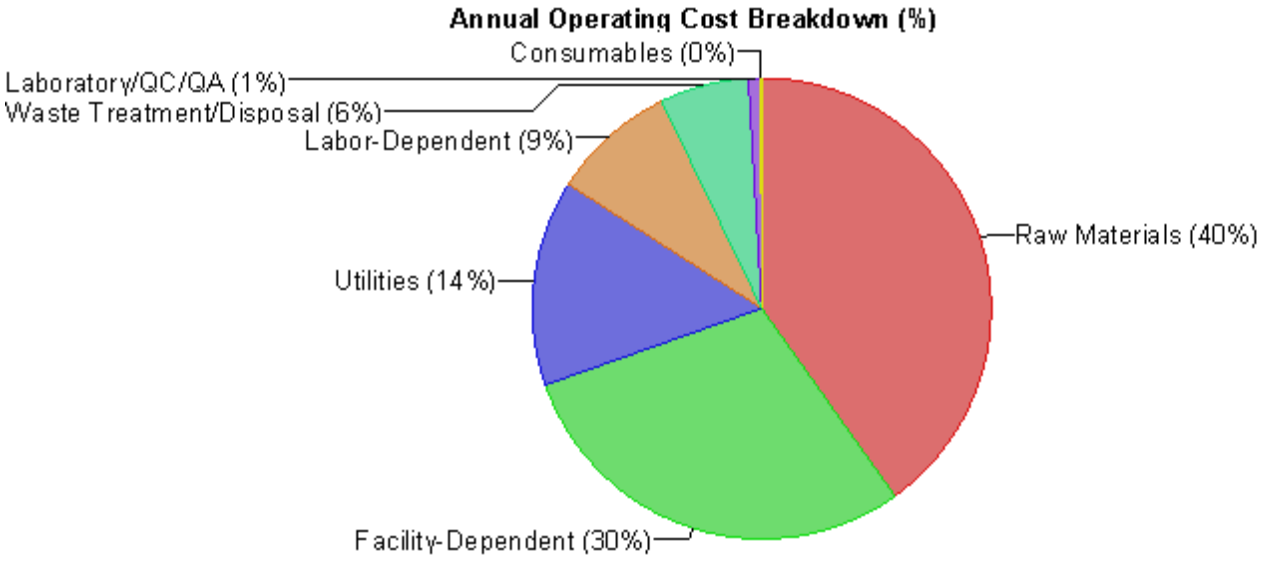



**Figure 4** Annual Operating Cost Breakdown

Based on the preliminary evaluation of this project's economics, one should not recommend investing in citric acid production unless there is a combination of favorable conditions. Obviously, availability of inexpensive equipment (e.g., by acquiring an existing facility) and raw materials (e.g., by locating the plant near a source of low cost molasses) are the most important factors. Development or adoption of a superior technology may also change the attractiveness of citric acid production. Such a technology is actually available and utilizes extraction for citric acid recovery [3]. Recovery by extraction eliminates the consumption of $Ca(OH)_2$ and $H_2SO_4$ and the generation of the unwanted $CaSO_4$. Butanol has been used as an extractant, as has tributyl phosphate. Ion pair extraction by means of secondary or tertiary amines dissolved in a water-immiscible solvent (e.g., octyl alcohol) provides an alternative route. With recent developments in electrodialysis membranes, the use of this technique to recover citric acid directly from the fermentation broth may become an attractive alternative [4].

# Miscellaneous Modeling Tips

## Equipment Sharing by Multiple Procedures

The  icon that appears on the lower left corner of a procedure icon indicates that multiple procedures are sharing the same equipment. For example, in this process it has been specified that procedures P-6

and P-8 share the same heat sterilizer (ST-101*).* The assignment of shared equipment between procedures is done by right-clicking on the procedure icon, selecting **Equipment Data** and choosing the equipment to be shared from the **Select** drop-down menu of the **Selection** frame on the upper left corner of the equipment data dialog.

## Continuous Unit Procedures in Batch Processes

SuperPro Designer handles modeling of batch processes, continuous processes, and combinations of batch and continuous. If a process involves several batch procedures that require inter-procedure scheduling links as well as a few continuous procedures, then the flowsheet's Mode of Operation should be set to "Batch" (under **Tasks➲ Set Mode of Operation**). This approach was followed for this example. If a mostly continuous process includes a few batch (cyclical) procedures that do not require inter-procedure scheduling, then the flowsheet's Mode of Operation should be set to "Continuous".

As mentioned earlier, the fermentation step (P-11) in this process is executed by a total of seven fermentors operating in Staggered Mode. This enables harvesting of a fermentation batch every 24 h. Please note that the SuperPro flowsheet displays the name of the first fermentor only (FR-101). However, all seven fermentors are displayed on the EOC. As noted previously, the equipment of a procedure is set to Stagger mode by right-clicking on its icon, selecting **Equipment Data**, checking the **On** box of the **Stagger Mode** frame, and specifying the number of extra equipment units.

The units of the isolation section of this process operate continuously. The V-105 storage tanks provide the buffer capacity between the two sections. Since the upstream section delivers a fermentation batch every 24 h, the continuous downstream purification equipment must be sized to process the material of one full fermentation batch every 24 h as well.

The mode of operation of a unit procedure is specified by right-clicking on it and selecting **Procedure Data**. That brings up the dialog of Figure 5. A procedure can be set to **Continuous** mode if its equipment is of continuous (throughput) nature and it includes only one operation. The process time of the single operation of a continuous procedure is set by SuperPro to be equal to the recipe cycle time (the time between consecutive batches) and cannot be edited by the user. Continuous procedures are ignored by default in scheduling calculations and their equipment occupancy is not displayed on the Equipment Occupancy Chart (EOC). However, if you check the "**Has Schedulable Start**" box, then continuous procedures are considered in scheduling and their equipment appears on the EOC.

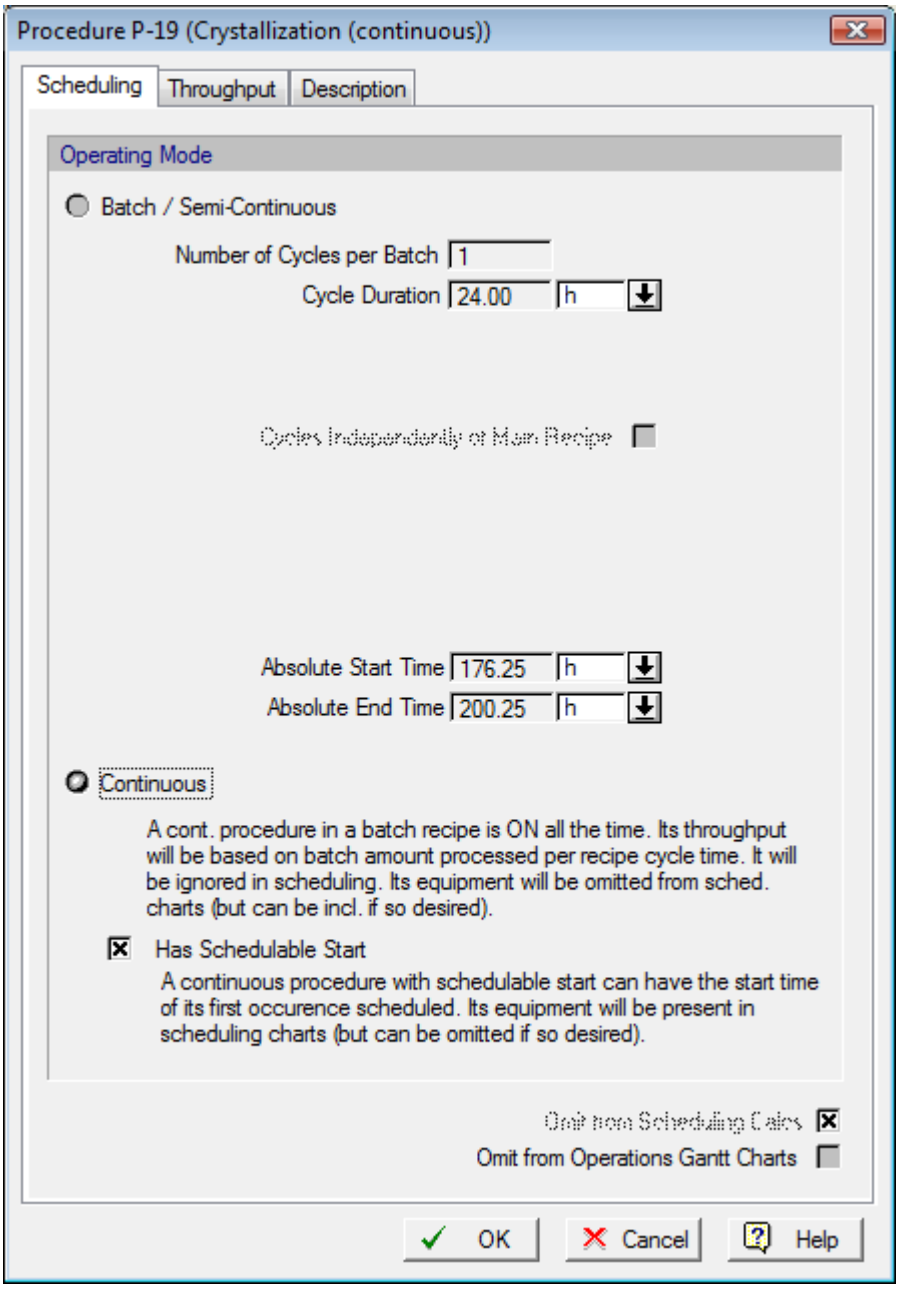



**Figure 5** The Procedure Data dialog for the Continuous Crystallizer (P-19/CR-101)

## Independently Cycling Unit Procedures in Batch Processes

If a batch (cyclical) procedure operates autonomously with its own rhythm (cycle time) that is not linked to the cycle time of the overall process, it is modeled in SuperPro by selecting the **Cycles Independently of Main Recipe** checkbox in the procedure data dialog (see **Figure 6**). The rotary vacuum filters (P-14 /

RVF-101, P-16 / RVF-102, P-18 / RVF-103, P-20 / RVF-104) in this process all operate in that mode. When the **Cycles Independently of Main Recipe** box is checked, a unit-procedure-specific cycle time may be specified. For instance, in the case of the P-14 rotary vacuum filter (RVF-101), the time between consecutive batch starts has been set as 31 h (Figure 6). Furthermore, the operations within this procedure have a total duration of 31 h (1 hour for applying the filter pre-coating plus 30 h of continuous filtration). The 31 h cycle time specified for this procedure represents the time between consecutive filtration cycles (batches), which is different from the overall recipe cycle time of 24 h. This way each fermentation batch is processed by 0.77 filtration cycles (batches). It is possible for the user to specify that ratio instead of the independent cycle time of an autonomous procedure (see Figure 6). A portion of the EOC which includes the autonomously-operating filters is shown in Figure 7.

**NOTES**

1) A clock icon (  ) is visible on the lower left side of each unit procedure that has been set to cycle independently.

2) Independently cycling procedures do not appear on the Gantt chart. However, their equipment occupancy is displayed on the EOC. Also, the operations of other procedures in the flowsheet cannot be scheduled relative to operations of independently cycling procedures.

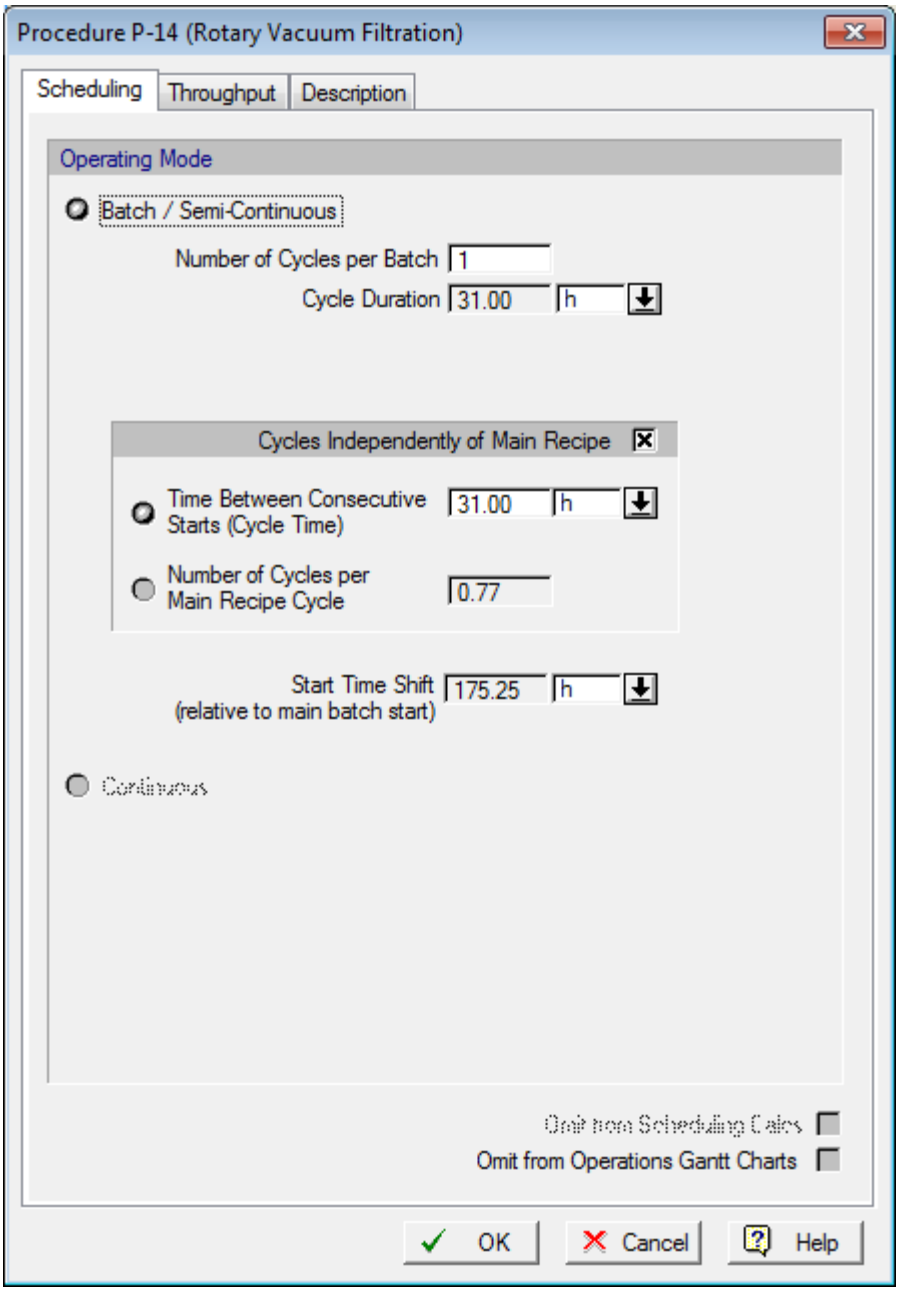



**Figure 6** The Independent Cycling specifications for the Biomass Removal filter (P-14/RVF-101)

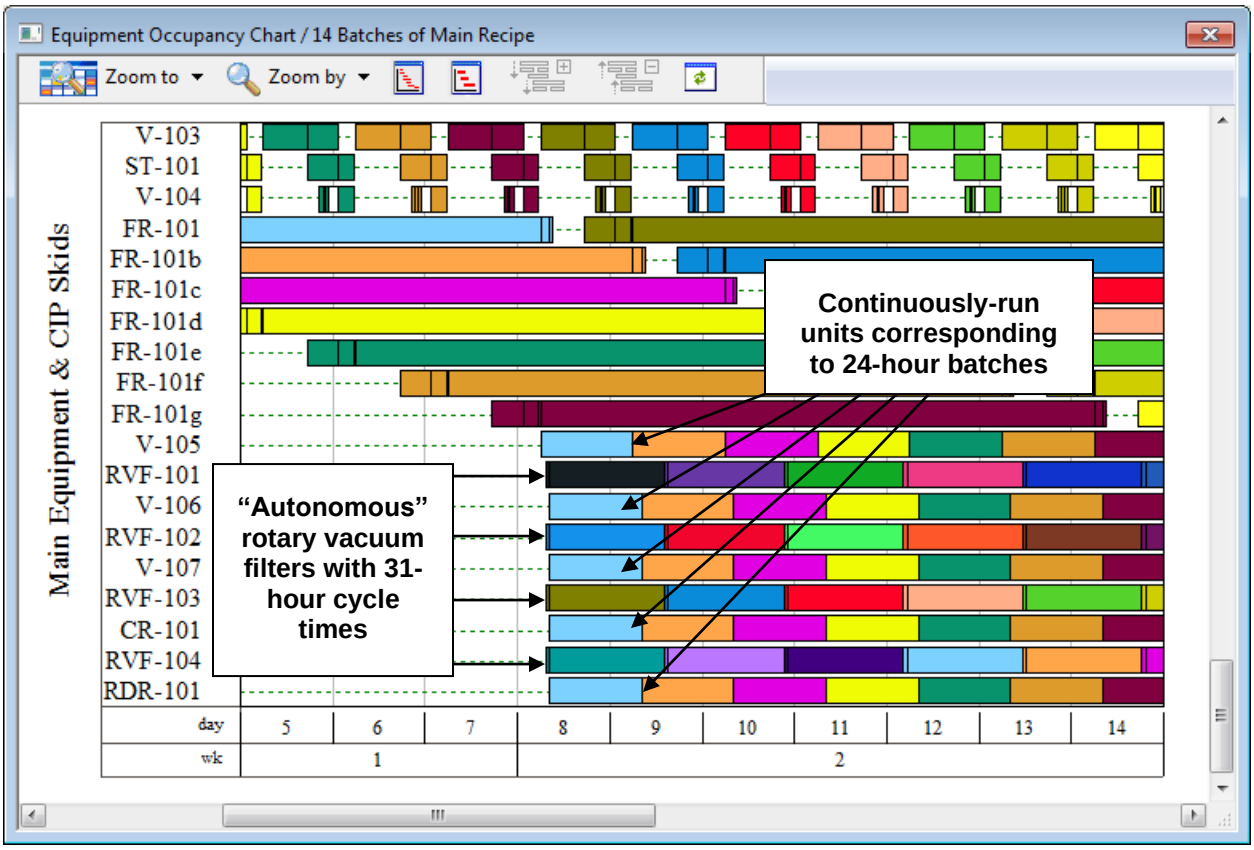



**Figure 7** Equipment occupancy of the autonomous procedures

## Pull-In Operations

A Pull-In operation can be used to intelligently charge material into a vessel by utilizing the **Advanced Options** of the **Amount** frame in the Oper.Cond's tab. Figure 7 displays the Advanced Options dialog of the Pull-In operation in P-1 that is used to dilute the molasses solution to 20% w/w Glucose content. In this case, the user simply specifies the composition of the diluant stream (Water-1b in this case) and SuperPro automatically calculates the amount of that stream so that the final Glucose concentration in the vessel is 20% w/w.

Pull-In operations also facilitate addition of material to achieve a certain final amount, a certain amount ratio (addition / initial), and to achieve a certain final temperature. The Custom-Mixer procedure (Unit Procedures➲ Mixing➲ Bulk Flow➲ Custom) offers similar functionality when dealing with streams instead of vessel contents.

Please note that a Pull-In operation can also function as a plain Charge operation when the amount (Mass of Volume) of the material to be charged is specified through the "Oper.Cond's" tab of the

operation's dialog (Amount frame). Furthermore, a Pull-In operation can function as a Transfer-In operation (to receive material from another procedure) when the "Available in Pull-In Stream" option is selected for Amount (see Figure 9).

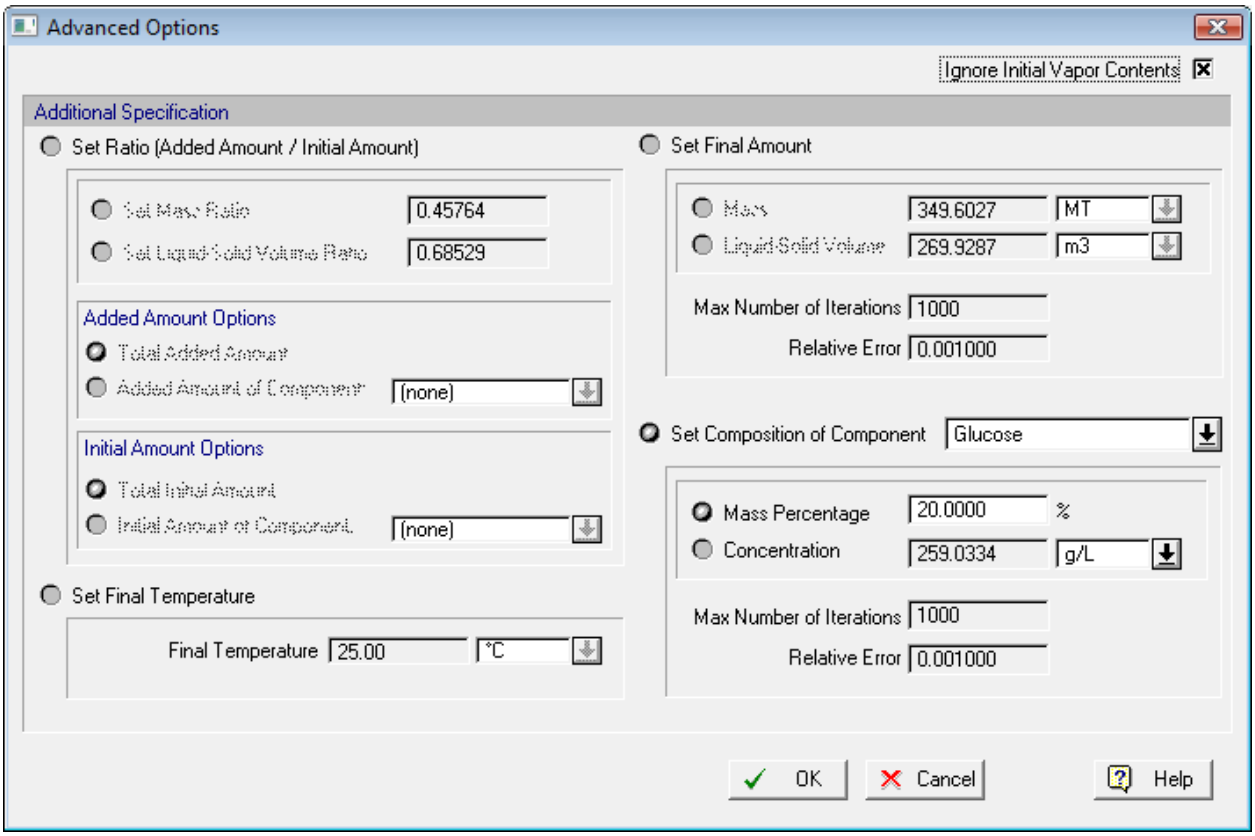



**Figure 7** The Pull-In Operation's Advanced Options Dialog

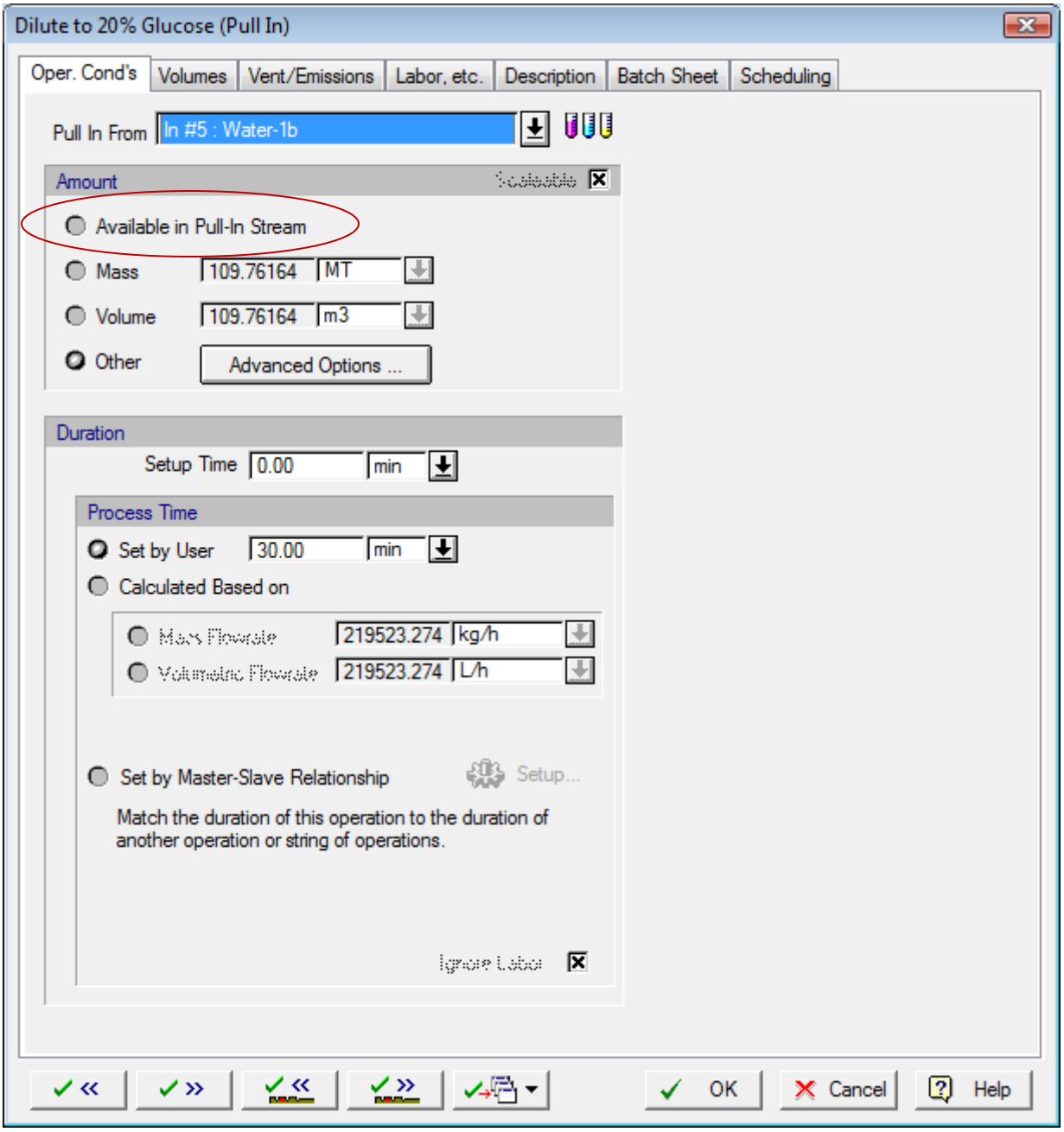



**Figure 8** Oper.Cond's tab of a Pull-In operation's dialog

## Specification of Excess Neutralizing (RxN) Agent

The neutralization operation of the Neutralization procedure (Unit Procedures➲ Continuous Reaction➲ Environmental Reaction➲ Stoichiometric➲ Neutralization) is modeled as a typical stoichiometric reaction with the additional functionality of being able to specify the **Excess (%)** of the neutralizing agent through

the Oper.Cond's tab of the operation's dialog (see Figure 10). Please note that the neutralizing agent must be a reactant (chemical component) present in the top inlet stream of the procedure (the Neutralizing Agent stream). SuperPro adjusts the flow of the Neutralizing Agent stream to satisfy the requirements of the reaction (stoichiometric amount plus excess amount). Product Precipitation in P-15 and Gypsum Formation in P-17 were modeled using Neutralization procedures. Ca-Hydroxide is the neutralizing agent in P-15 (5% excess) and Sulfuric Acid is the neutralizing agent in P-17 (4% excess). Sulfuric Acid is added as a 10% w/w sulfuric acid solution in water.

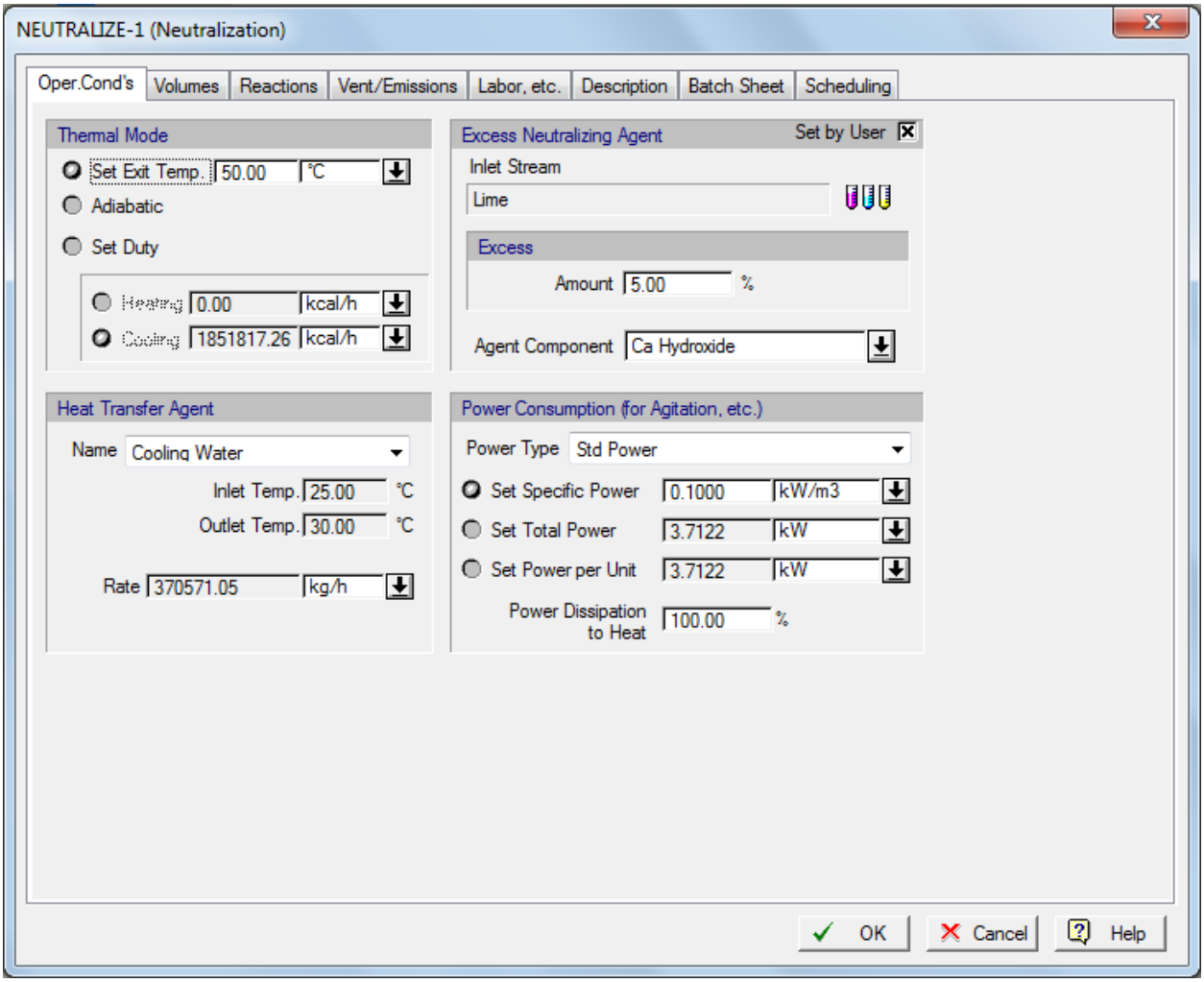



**Figure 10** Oper.Cond's tab of a Neutralization operation's dialog window.

## User Defined Equipment Cost Models

SuperPro Designer facilitates estimation of capital and operating costs and it also performs thorough project economic evaluation, profitability analysis and cash flow analysis of integrated manufacturing and environmental processes. The purchase cost of equipment is an important parameter that affects the direct fixed capital investment of a project and indirectly affects the operating cost. SuperPro is equipped

with correlations for estimating the purchase cost of equipment. However, the built-in correlations for most types of vessels, membrane filters, chromatography columns and various other types of equipment are more suitable for fine chemical and pharmaceutical types of facilities than for commodity chemicals such as citric acid.

For commodity bio/chemicals and bio-fuel processes, we advise users to enter their own equipment cost data for better accuracy. A good source of equipment cost data for such processes is available from the US Department of Energy [5].

Figure 11 displays the **Purchase Cost** tab of the equipment data dialog of a unit procedure (fermentor FR-101 in this case). This tab can be accessed by right clicking on the procedure's icon, selecting **Equipment Data**, and then switching to the **Purchase Cost** tab. As you can see, the cost estimation options include a user-defined purchase cost, estimation through the built-in cost model, and a user-defined cost model.

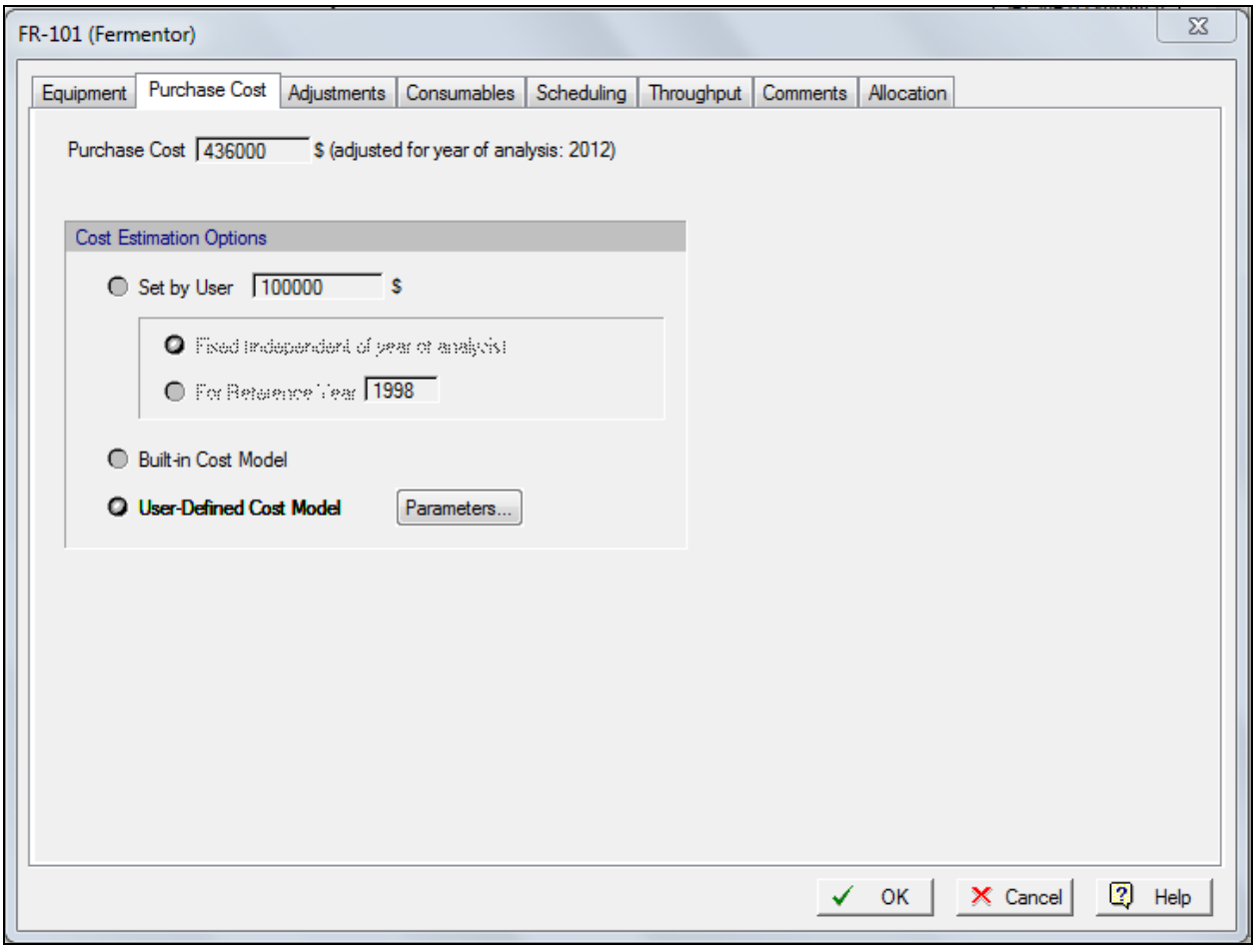



**Figure 11** Equipment Purchase Cost Tab

A new cost model can be specified by selecting the **User-Defined Cost Model** option and clicking on the **Parameters** button. This brings up the dialog shown in Figure 12, through which the user can specify cost correlations in a power-law format (i.e., the red equation at the top of the window). The parameters of the equation are the following:

- $C_0$ is the base cost,
- $Q_0$ is the base capacity, and
- Q is the actual capacity, and
- a is the exponent of the power law

The capacity range can be broken down into any number of intervals.

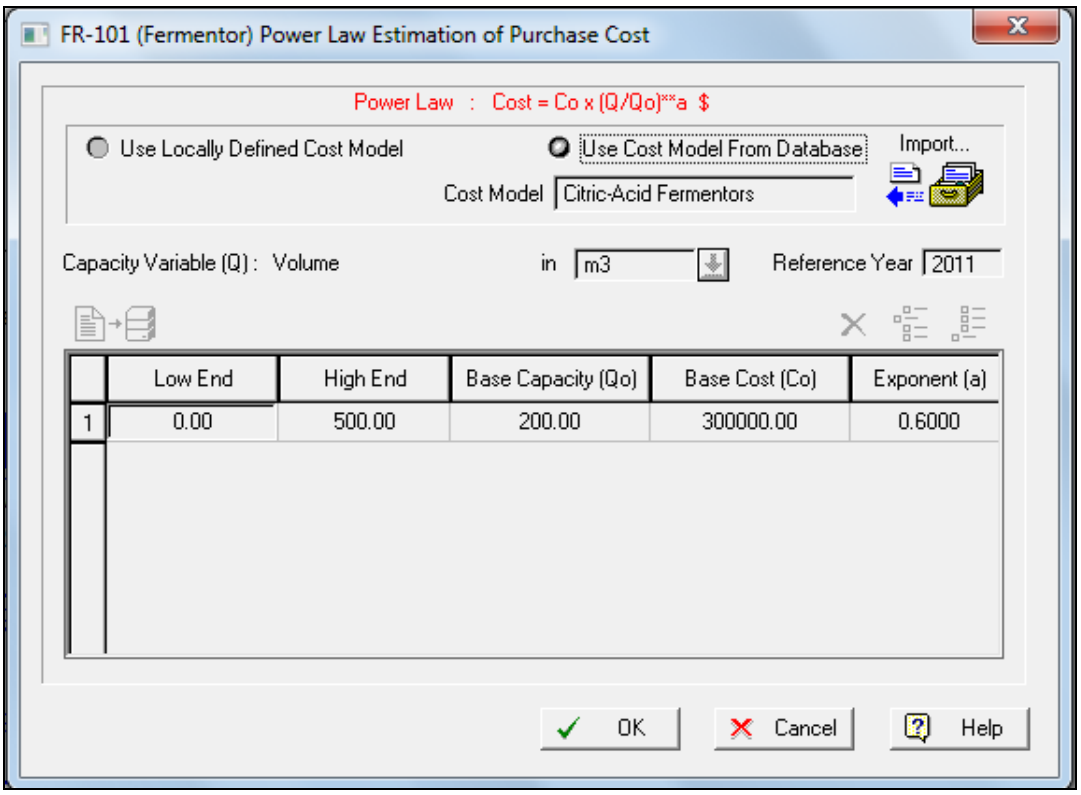



**Figure 12** User Defined Equipment Cost Correlation Dialog

Furthermore, users have the option to define the cost model locally into the file or retrieve it from the equipment cost database associated with their particular installation. To specify the model locally, select the **Use Locally Defined Cost Model** option in Figure 12. Select the **Use Cost Model from Database** option for retrieving a cost model from the user database. The latter option requires that the user defines the cost model in the user databank before it can be retrieved. This option is grayed out if no compatible equipment cost data model exists in the database. An advantage of defining the cost model in the database is that the data can be retrieved by multiple equipment items of that type, which may exist in multiple SuperPro files developed by various SuperPro users. As a result, the use of the user cost model

database can ensure consistency in equipment cost calculations across many projects. For additional information on the user databanks of SuperPro, please consult the corresponding sections of the SuperPro manual and help facility, as well as the SynPharmDB ReadMe file of the SynPharm example (one of the examples that are shipped with SuperPro).

To define the equipment cost model in the user database, select **Databanks➲ Equipment➲ User Defined Cost Models** from the main menu. This displays the window shown in Figure 13. Next select the appropriate **Equipment Type** through the drop-down menu at the top and click on the **Add New** button (  ). This brings up the dialog shown in Figure 14 through which a user cost model can be defined. Finally, to access the newly created model and add its data to an equipment item, revisit the window of Figure 12 and click on the **Import** button (upper right corner of the dialog). Please note that it is also possible to first add the model locally to a file and then deposit it into the user database.

SuperPro is also equipped with databases for storing cost and other data for equipment of specific sizes that can be purchased from various vendors. That data becomes accessible to a piece of equipment in a file by matching it with suitable equipment in the DB through the **Allocation** tab of the **Equipment Data** dialog of an equipment item. For additional information on this subject, please consult the corresponding sections of the SuperPro manual and help facility, as well as the SynPharmDB ReadMe file of the SynPharm example.

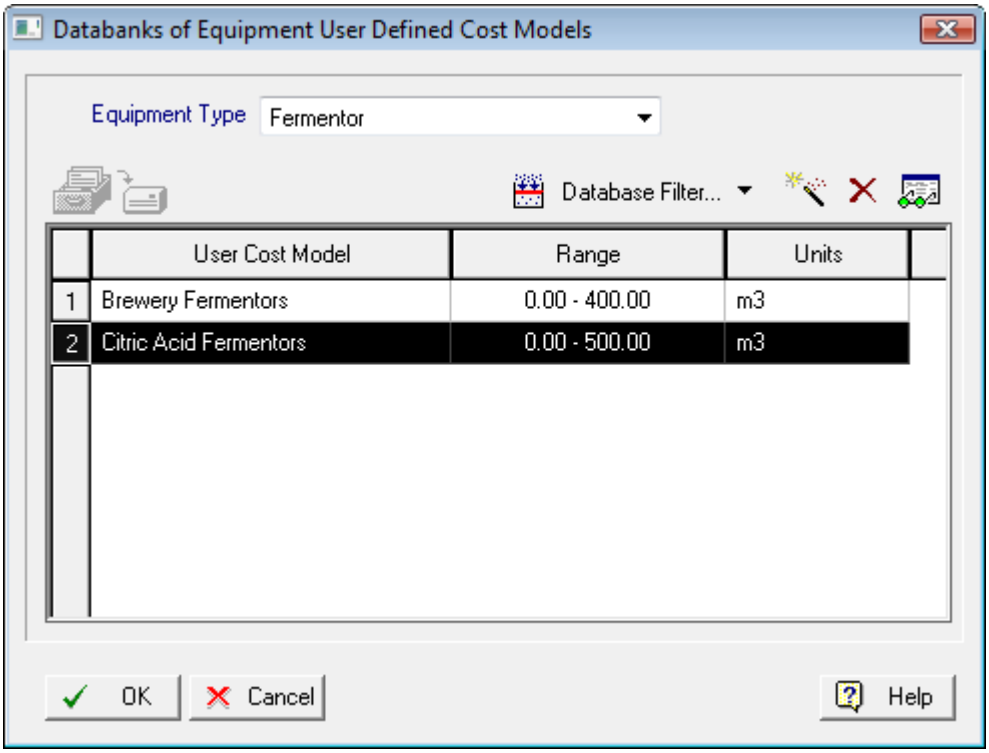



**Figure 13** Equipment User Defined Cost Model Databank Window

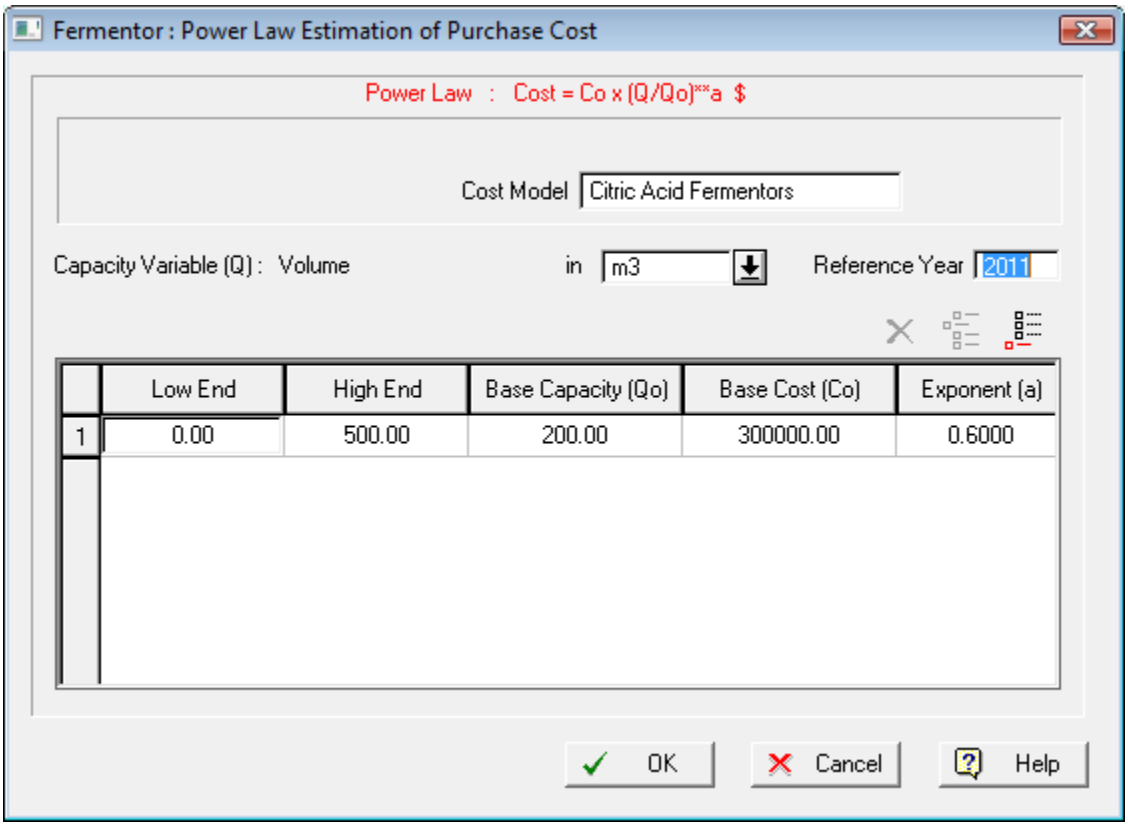



**Figure 14** User Cost Model Definition Dialog

## Material Storage Units

Material storage units in SuperPro Designer may be thought of as *virtual tanks* (i.e., in the sense that they are not displayed in the flowsheet) which handle the supply or receiving of materials during a process. A receiving storage unit with the name **Liquid Waste Tank** was added to this example in order to collect all liquid waste streams generated throughout the process. Waste generated during the CIP operation of the fermentor (P-11) was also assigned to be collected in this storage unit.

Storage units can be added to a SuperPro file by selecting **Tasks➲ Other Resources➲ Material Storage Units** and clicking on the *Add New Receiving Storage Unit* button (i.e., the button with the magic wand icon). Selecting a storage unit and clicking on the spectacles button ( ) brings up a dialog through which the user can view and modify its properties. The properties dialog includes several tabs. The first is the *Properties* tab through which the user can specify the name of the unit, classify the type of material in the unit, and provide other information such as the cost for waste treatment. The second tab

corresponds to the *Availability Limits* of the storage unit. The third tab is the *Inventory Data* tab and the final tab is the *Locations* tab which displays the operations and streams that utilize the unit.

To assign an output stream to a receiving storage unit, right click on the stream to access the context menu, select the **Assign Receiving Storage Unit** option, and select the desired storage unit from the drop down menu. Furthermore, to send the output material of a CIP operation to a receiving unit, open its operation dialog, select a cleaning step from the **Cleaning Step Sequence** list (on the right) and then select the desired storage unit from the drop down list in the Waste frame on the bottom of the dialog (see Figure 15). When a receiving tank is selected, the waste is automatically classified according to the type of waste which has been defined for it (e.g. aqueous waste, organic waste, etc.). If the *Remove Contents as Waste with Disposed Agent* box is checked for a CIP step, any material present in the equipment prior to the CIP operation is removed as CIP waste.

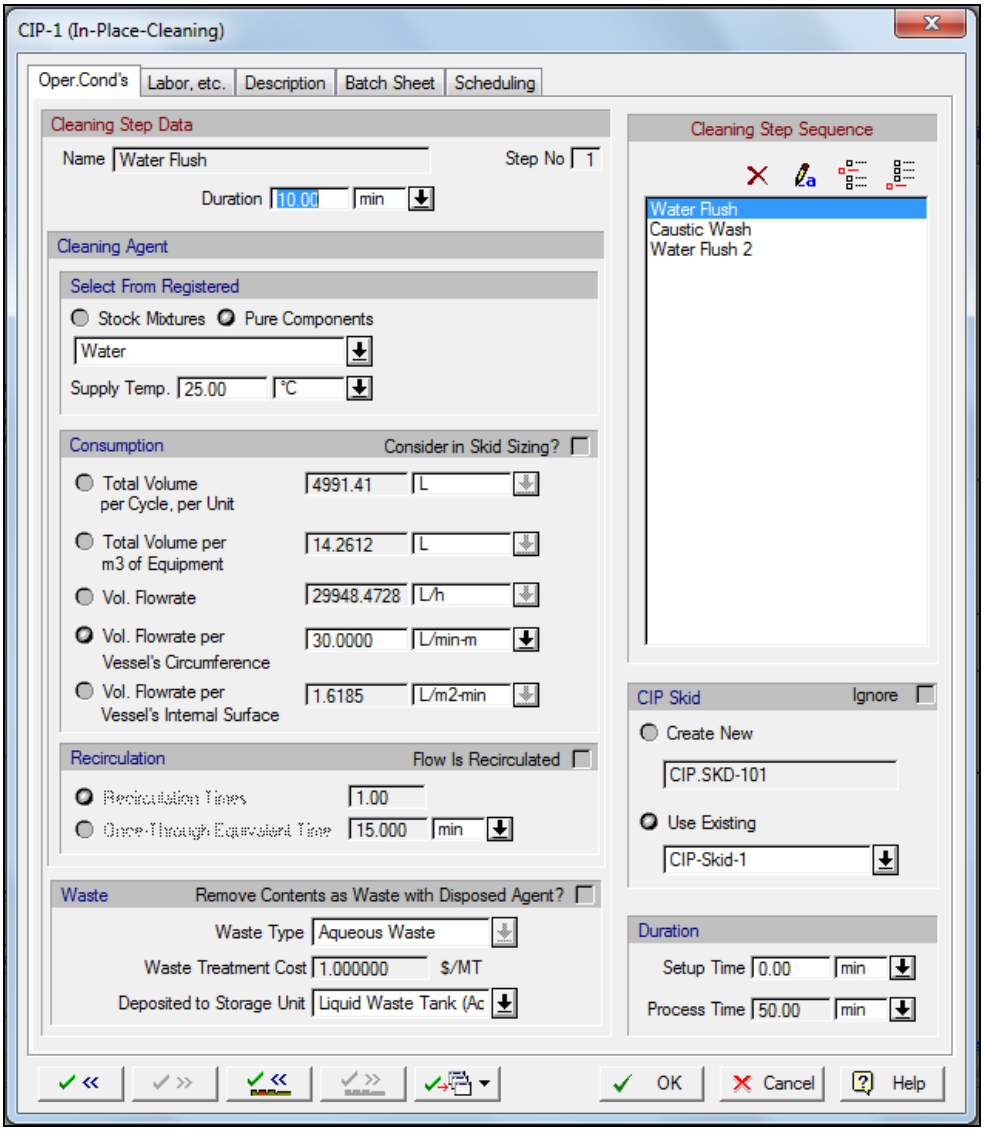



**Figure 15** CIP Operating Conditions Dialog

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# Citric Acid Production

